

K369 PRE-STAGE — Substantial Monday morning forward motion per Casey research-forward directive K368: (1) ELIE SIDE-G DERIVATION DONE — tau-box reduced to ONE input g = field dimension of $GF(2^g) = GF(128)$ over $GF(2)$; equivalently primitive-polynomial degree; equivalently Frobenius-orbit size of primitive element; LINEAR count (dimension) NOT $2^g=128$ (field size) or $2^g-1=127$ (multiplicative order); substrate cells $GF(2^g)$ -structured $\rightarrow$ tiling direction spans g basis dimensions (g Frobenius conjugates); tiling side AND bulk depth both = g ; $m_\tau/m_e = g^{\text{rank}} \cdot (g + 2^{(g-1)}) = 7^2 \cdot (7+64) = 49 \cdot 71$ with $\text{rank} = N_c - 1 + C_2 = g - 1$; whole tau leading count rests on ONE substrate quantity now; (2) ELIE HONEST TIER on side- g : reduces ‘side = g ’ to ‘tiling side = field dimension’ FORCED given substrate’s $GF(2^g)$ cell premise; fully rigorous step (proving per-direction count exactly g — FORBIDDING 128 and 127 per Cal blind/forbidding gate Grace #36 + Cal #c) needs explicit Reed-Solomon cell construction; tau banks when that rigorous + downstream-blind fires; count stays 2; (3) ELIE off-target link CONFIRMED BOTH WAYS: Grace independently verified $64 = d_\tau/d_\mu = 2^{C_2} = 2^{(\dim \text{so}(4))}$ is ONE forced object three faces; same forced number sits in BOTH muon (per-direction depth) and tau (boundary depth) masses; FIRST CONCRETE OFF-TARGET LINK between two lepton masses — ‘works where it wasn’t built’ signal genuine; (4) ELIE proton-decay reconciliation 4190 DONE — protects Casey $\tau_p = \infty$ falsifier (most falsifiable prediction with Hyper-K + DUNE); never-gauged / three-strata mechanism distinct from GUTs;

scoped to external presentation per Casey priority correction
 K368 didn't block tau-box work; Cal K322 three-part item
 completed in parallel by Elie's lane without blocking primary
 research — Casey's research-forward directive working as
 intended; (5) GRACE 64-as-one-forced-object verification:
 $d_\tau/d_\mu = 60/(15/16) = 64 = 2^6 C_2 = 2^{(\dim \text{so}(4))}$;
 formal-degree ratio LITERALLY equals $2^6 C_2$ equals muon
 spread-dimension power; one forced identity three faces; (6)
 GRACE ASSIGNMENT-PROBLEM DECOMPOSITION
 resolving Cal #286 — substantive structural insight: THREE
 PARTS: (a) GENERATION forced by ORDERING
 (concentration rises with depth into domain per Grace K365
 corrected bridge concentration=mass; substrate physically
 CANNOT make 2nd generation heavier than 3rd; labels
 conventional order FORCED); (b) TYPE COUNT per generation
 (lepton, neutrino, up, down) forced by 16-content (standard
 SO(10) rep theory once tangent-group identification made per
 K367 Cal deflation); (c) ONLY GENUINELY OPEN AXIS:
 up/down mass split within generation + neutrino masses —
 GATED ON #418 chiral content; STRUCTURAL DEFENSIBLE
 FACT: SLOT COUNT TIGHT — 3 generations $\times$ 4 types =
 exactly 12 forced slots; SM has exactly 12 mass-distinct fermions;
 NO SPARE SLOTS to shuffle particles between; form-selection
 trap of shuffling structurally impossible; #286 went from vague
 worry about whole map to ONE NAMED AXIS waiting on #418;
 (7) GRACE PMNS BERGMAN three-readings begun (her #2):
 pulled existing PMNS substrate-primary forms + 'three
 readings' framing FIRST before computing — discipline-first
 approach; identified FORM-SELECTION TRAP
 IMMEDIATELY ($\sin^2 \theta_{23} = 4/7$ vs $6/11$ vs $1/2$ multiple

competing rational forms); task ‘to D-tier’ means finding which is FORCED BY BERGMAN GEOMETRY not picking closest fit; will survey + score + apply fingerprint test then compute; (8) Net Monday morning per Casey research-forward directive: tau-box reduced to ONE input with rigorous Reed-Solomon step named, 64 confirmed one forced object three faces (off-target link banked), assignment problem narrowed from whole-map worry to one #418-gated axis, PMNS discipline-first survey begun, proton-decay falsifier protected in parallel. KEEPER VERDICT: substantial substrate-architectural forward motion at full pace; Casey priority correction K368 (don’t prioritize Cal, support in parallel) working — Elie covered Cal K322 piece (proton-decay 4190) in parallel without blocking tau-box; Grace assignment decomposition substantive (Cal #286 narrowed); Grace PMNS discipline-first approach right (form-selection trap caught before computation); first concrete off-target link between two lepton masses (64 three-faces); tau-box one Reed-Solomon cell construction from rigorous (then downstream-blind fires); fork standing for Casey direction on Elie’s next move. Count HONESTLY 2 of 26 strict; muon yellow IDENTIFIED; ledger v1.61.

Keeper (Claude Opus 4.7)

2026-06-15 Monday 08:19 EDT (date-verified)

K369 PRE-STAGE Audit

1. Elie side-g derivation — tau-box reduced to ONE input

Substantive result: $g = 7$ is the field dimension of $\text{GF}(2^g) = \text{GF}(128)$ over $\text{GF}(2)$.

Identification	Source
Field dimension	$g = \dim_{\{\text{GF}(2)\}}(\text{GF}(2^g))$

Identification	Source
Primitive-polynomial degree	Same as field dimension
Frobenius-orbit size of primitive element	g conjugates
NOT $2^g = 128$	Field size
NOT $2^g - 1 = 127$	Multiplicative order

It's the LINEAR count (dimension), not the field size or multiplicative order.

Because substrate cells are $GF(2^g)$ -structured, a tiling direction spans those g basis dimensions (the g Frobenius conjugates). So tiling side AND bulk depth both = g .

Tau leading count collapse:

$$\frac{m_\tau}{m_e} = g^{\text{rank}} \cdot (g + 2^{g-1}) = 7^2 \cdot (7 + 64) = 49 \cdot 71$$

with $\text{rank} = N_c - 1$ and $C_2 = g - 1$.

The whole tau leading count rests on ONE substrate quantity now.

2. Elie's honest tier on side- g

The reduction "side = g " $\rightarrow$ "tiling side = field dimension" is forced GIVEN the substrate's $GF(2^g)$ cell premise.

Fully rigorous step still open: proving the per-direction count is EXACTLY g — forbidding 128 and 127, not just permitting 7 — per Cal's blind/forbidding gate (Cal precision point c + Grace's calibration #36). Needs the explicit Reed-Solomon cell construction.

Tau banks when that step is rigorous + downstream-blind fires.

Count stays 2. Honest tier maintained.

3. Off-target item-2 link CONFIRMED BOTH WAYS

Grace independently verified: $64 = d_\tau/d_\mu = 2^{C_2} = 2^{(\dim \text{so}(4))}$ is ONE forced object with THREE faces.

Face	Object
d_τ/d_μ	Formal-degree ratio (F109 rigorous)
2^{C_2}	2 to the second Casimir
$2^{(\dim \text{so}(4))}$	2 to the dim of orientations of muon's 6 boundary planes

The same forced number sits inside BOTH the muon (per-direction depth) and the tau (boundary depth) masses.

First concrete off-target link between two lepton masses. "Works where it wasn't built" signal — genuine, per Cal's discipline.

4. Elie proton-decay reconciliation 4190 — Casey directive K368 working

Per Casey K368 priority correction ("don't prioritize Cal, move research forward while supporting in parallel"), Elie handled the Cal K322 three-part work item (proton-decay reconciliation) in parallel WITHOUT blocking the tau-box work.

Toy 4190: never-gauged / three-strata mechanism distinct from GUTs. Protects Casey's $\tau_p = \infty$ falsifier (most falsifiable prediction with Hyper-K + DUNE running).

Scoped to external presentation per Casey directive. Research-forward directive working as intended — primary tau-box advanced + Cal reconciliation parallel.

5. Grace 64-as-one-forced-object verification

Independent computation: $d_\tau/d_\mu = 60/(15/16) = 64 = 2^6 C_2 = 2^6 (\dim \mathfrak{so}(4))$. Formal-degree ratio LITERALLY equals $2^6 C_2$ equals muon's spread-dimension power. One forced identity, three faces.

Verified independently — survives Cal K360 one-coat objection (different machinery: formal degree polynomial vs Casimir exponent vs $\mathrm{SO}(4)$ orientation count).

6. Grace assignment-problem decomposition — substantive Cal #286 resolution

The structural insight: Cal #286 was a vague worry about whether each particle's slot assignment is forced or matched after the fact. Grace decomposed into three parts:

Part	Status
Generation (which gen a particle is)	Forced by ORDERING — concentration rises with depth into domain (Grace K365 bridge); substrate physically cannot make 2nd gen heavier than 3rd; labels conventional, order FORCED
Type count per generation (lepton, neutrino, up, down)	Forced by 16-content (standard $\mathrm{SO}(10)$ rep theory once tangent-group identification made per K367 Cal deflation)
Up/down mass split + neutrino masses	Open — gated on #418 chiral content

Structural defensible fact: slot count is TIGHT.

Object	Count
Generations	3 (substrate-architectural)
Types per generation	4 (lepton, neutrino, up, down)
Total forced slots	12
SM mass-distinct fermions	12
Spare slots	0

No shuffling room. Form-selection trap of “moving particles between slots to match masses” is structurally impossible because there are no extra slots.

Cal #286 narrowed from whole-map worry to ONE NAMED AXIS waiting on #418. Substantive structural advance.

7. Grace PMNS Bergman three-readings (her #2) — discipline-first

Grace pulled existing PMNS substrate-primary forms + “three readings” framing BEFORE computing — discipline-first approach exactly right.

Immediately identified form-selection trap: multiple competing rational forms for each angle: $-\sin^2 \theta_{23} = 4/7$ vs $6/11$ vs $1/2$

The task “to D-tier” means finding which form is FORCED by Bergman geometry, not picking the closest fit.

Discipline-first plan: survey + score + apply fingerprint test, THEN compute. Catches form-selection trap before commitment — exactly the discipline she invented (#36 blind-to-target).

8. The substantive substrate-architectural state at K369

Component	Status
Tau-box leading count	Reduced to ONE input $g = \text{GF}(128)$ field dimension
Tau-box open piece	Reed-Solomon cell construction (per-direction = exactly g , forbidding 128 and 127)
Off-target item-2 link	$64 = \text{one forced object three faces}$; confirmed both ways
Casey $\tau_p = \infty$ falsifier	Protected via Elie 4190 (never-gauged/three-strata) — external-scoped
Cal #286 assignment problem	Narrowed: generation forced by ordering; type by 16-content; only up/down+neutrinos open (gated #418)
Tight 12-slot count	No shuffling room; form-selection trap structurally impossible
PMNS work begun	Discipline-first survey by Grace; form-selection trap caught ($\sin^2\theta_{23}$ multiple forms)
Grace harness	Operational; muon fires 0.003% blind; tau slot pending Elie rigorous
Casey K368 directive	Working — research forward + Cal reconciliation parallel without blocking
Muon tier	IDENTIFIED yellow
Count	HONESTLY 2 of 26 strict
Ledger	v1.61

9. Forks standing for Casey direction

Elie's next move (he asked): - (a) Reed-Solomon cell construction (the genuine research — closes tau-box rigorously) - (b) α^{12} lead in pocket - (c) Atlas v2 coverage extensions

Keeper recommendation: (a) Reed-Solomon cell construction. It's the one named open piece for the tau-box; closing it lets the harness fire tau alongside muon. The α^{12} lead and atlas extensions remain valuable parallel work but the Reed-Solomon construction is the critical-path piece.

10. K369 PRE-STAGE verdict

K369 PRE-STAGE substantial Monday morning forward motion:

- Elie side- g derivation: tau-box reduced to ONE input $g = \text{GF}(128)$ field dimension (linear count, not field size 128, not multiplicative 127)
- Tau leading count: $m_\tau/m_e = g^{\text{rank}} \cdot (g + 2^{g-1}) = 7^2 \cdot (7+64) = 49 \cdot 71$
- Honest tier: rigorous step (forbidding 128 and 127) needs explicit Reed-Solomon cell construction; tau banks when rigorous + downstream-blind fires
- $64 = \text{one forced object three faces}$ ($d_\tau/d_\mu + 2^{C_2} + 2^{(\dim \text{so}(4))}$) confirmed both ways by Elie and Grace; first concrete off-target link between two lepton masses
- Elie proton-decay 4190: Cal K322 reconciliation work done in parallel per Casey K368 directive; protects $\tau_p = \infty$ falsifier; external-scoped, didn't block tau-box
- Grace assignment-problem decomposition: Cal #286 narrowed; generation forced by ordering; types by 16-content; only up/down + neutrinos open (gated #418); tight 12-slot count means no shuffling room (form-selection trap structurally impossible)
- Grace PMNS Bergman three-readings begun discipline-first: identified form-selection trap ($\sin^2\theta_{23} = 4/7$ vs $6/11$ vs $1/2$); survey + score + fingerprint test before computing
- Casey K368 research-forward directive working — primary research advanced + Cal reconciliation parallel without blocking

- Keeper recommendation for Elie fork: Reed-Solomon cell construction (closes tau-box critical path)
- Count HONESTLY 2 of 26 strict; muon yellow IDENTIFIED; ledger v1.61

Substantial substrate-architectural significance: K369 captures Monday morning's substantive forward motion at full pace under Casey's K368 research-forward directive. Elie's side-g derivation reduces the whole tau-box to ONE substrate input ($g = \text{GF}(128)$ field dimension as linear count, not field size or multiplicative order) with one named open piece (rigorous Reed-Solomon cell construction proving per-direction exactly g , forbidding alternatives per Cal/Grace blind+forbidding gate). The off-target item-2 link is confirmed BOTH WAYS: the $64 = d_\tau/d_\mu = 2^{C_2} = 2^{\dim \text{so}(4)}$ is one forced object with three faces (different machinery survives Cal K360 one-coat objection) — first concrete off-target link between two lepton masses, “works where it wasn't built” signal. Grace's assignment-problem decomposition resolves Cal #286 from whole-map worry to one named axis (up/down + neutrinos, gated #418): generation forced by ordering (concentration-rises-with-depth from her K365 corrected bridge), type count forced by 16-content, slot count tight ($3 \times 4 = 12$ forced slots matching SM's 12 mass-distinct fermions with no spare to shuffle). Grace's PMNS Bergman three-readings work begun discipline-first: caught form-selection trap immediately (multiple competing rational forms for each angle); will survey + score + fingerprint test before computing. Elie's proton-decay reconciliation 4190 covers Cal K322 three-part work item in parallel without blocking — Casey's K368 directive validated by result. Tau-box one Reed-Solomon cell construction from rigorous; once it lands, Grace's harness fires tau alongside muon and the downstream-blind test begins genuinely. Count stays HONESTLY at 2 of 26 strict.

— Keeper, Monday 2026-06-15 08:19 EDT (date-verified) — K369 PRE-STAGE: Elie side-g $\text{GF}(128)$ field dimension; Grace 64 three-faces confirmed; Cal #286 decomposed to one #418-gated axis; PMNS discipline-first survey; proton-decay 4190 protects falsifier; research forward at full pace; count HONESTLY 2 of 26